

Harshith (Harry) Bhattaram

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SUMMARY

Data Scientist with 2+ years shipping ML systems in production, including a real-time computer vision system to run inference on live fleet vehicles. Currently researching multimodal RAG for clinical applications at University at Buffalo, building a retrieval pipeline over 250k+ medical image-report pairs to answer queries under 1 second. Comfortable across the full stack, building GBM/LSTM forecasting models on telematics data and explainability into every deployment (SHAP, TreeExplainer).

EDUCATION

Master of Science, Computer Science | University at Buffalo, The State University of New York, USA

WORK EXPERIENCE

University at Buffalo

Buffalo, NY

Machine Learning Research Assistant

March 2026-Present

- Built a Clinical Radiology Report Generation via Multimodal Retrieval-Augmented Generation (RAG).
- Architected a multimodal RAG pipeline on the MIMIC-CXR dataset (250k+ chest X-ray image-report pairs).
- Used BioMedCLIP (ViT-B/16) for cross-modal embeddings, retrieval via FAISS indexing, dual layer generation module.
- Orchestrated a memory-constrained subprocess architecture to get <1second query latency on knowledge base.
- Evaluated with CheXbert, BLEU, ROUGE-L, BERTScore to quantify accuracy against ground-truth reports.

LocoNav Fleet Management Solutions

Gurugram, India

Software Engineer (Data Scientist)

Jun 2022 - Jun 2024

- Built a driver drowsiness and intoxication classifier using OpenCV, Tensorflow on 10k+ video frames, achieving 87% accuracy.
- Reduced false negative drowsiness detections by 25% by incorporating facial landmark tracking and yawn angle-based alertness.
- Cut model inference response latency by 20% via optimized production pipeline that triggered real-time low-latency in-app alerts.
- Developed GBM, LSTM ETA prediction models on GPS, telematics data across 50k+ trips, improved forecast accuracy by 15%.
- Reduced average trip duration by 12% via graph-based route optimization using real-time traffic features, historical travel data.
- Collaborated with cross-functional teams in Agile workflows, supporting planning, stand-ups, and retrospectives.

Software Engineer (Data Science Internship)

Jan 2022 - May 2022

- Built a predictive maintenance pipeline from time-series telemetry data, handled class imbalance in failure labels via SMOTE.
- Achieved 85% precision in component failure prediction using ensemble classifiers with engineered degradation features.
- Reduced unplanned vehicle downtime by 18%, deployed automated ML alert pipeline integrated with fleet operations dashboard.

PROJECTS

Credit Card Customer Churn Prediction System with Explainable AI Dashboard | [Link](#)

- Built end-to-end ML pipeline using XGBoost and Logistic Regression on 10k+ customers on kaggle credit card dataset.
- Achieved ROC-AUC of 0.99 and 90.5% recall after engineering on 21 features and validating feature design with TreeExplainer.
- Deployed an interactive Streamlit application which includes EDA, model performance, and churn probability per customer.
- Integrated SHAP explainability for global feature importance & waterfall plots for model interpretability to show key drivers.

Healthcare Patient Readmission Risk Modeling with Explainability Report | [Link](#)

- Analyzed the UCI diabetes readmission dataset (100k+ rows) with SQL profiling queries to find readmission rates across cohorts.
- Engineered 44 features, resolved class imbalance using XGBoost instead of SMOTE to avoid interpolation errors on mixed data.
- Trained a gradient boosted classifier, achieved statistical improvements, validated 11 risk factors via chi-square & Welch's t-tests.
- Autogenerated clinical decision report with SHAP waterfall charts, confusion matrix, feature importance and deployed on cloud.

NLP Sentiment Analysis & Topic Modeling Platform for Product Reviews | [Link](#)

- Designed a full NLP pipeline processing 24k + Amazon reviews via VADER and DistilBERT sentiment analysis, LDA modeling.
- Engineered modular text preprocessing via spaCy lemmatization & custom cleaning logic, parquet schema with 20 columns.
- Reduced 27+ raw records to 24k analysis ready rows, tracked sentiment scores, topic labels & model agreement via NLP model.
- Deployed a 4-page interactive Plotly dashboard on streamlit with time-series trend analysis by sentiment, visualizations per topic.

SKILLS

- **Languages and Tools:** Python, SQL, PostgreSQL, R, Git, Tableau, Plotly, Streamlit, Render
- **Data Handling:** Data Cleaning, Data Preprocessing, Structured data, Unstructured data, EDA, Feature engineering
- **Machine Learning & Statistics:** Scikit-Learn, XGBoost, Gradient Boost, Random Forest, LSTM, ARIMA, Transformers, Numpy, Keras, Cross-Validation, Hypothesis Testing, SMOTE, PCA, Pandas, Time-Series Analysis, Matplotlib
- **Deep Learning & NLP:** TensorFlow, PyTorch, DistilBERT, BioMedCLIP, OpenCV, BERTopic, TF-IDF, LDA, Named Entity Recognition (NER), SHAP, LIME, Hugging Face,
- **Generative AI:** RAG, FAISS, Prompt Engineering, QLoRA, LangChain, LangGraph, LLMs VLMs, Fine-tuning
- **Evaluation & Cloud:** ROC-AUC, ROUGE, BLEU, BERTScore, FastAPI, Docker, AWS (S3, Lambda, RDS), Azure, GCP